

Selective Prediction and the Persistence Illusion: A Diagnostic Decomposition of VIX Regime Classification

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Abstract: Confidence-based selective prediction systems for financial regime classification can achieve apparent accuracy exceeding 90% while providing no genuine predictive skill beyond naive persistence. We demonstrate this through a systematic diagnostic study of VIX regime classification across 5,000 trading days (July 2006–May 2026). A six-component evaluation protocol—comprising coverage-matched baseline comparison, persistence quantification, regime-transition accuracy audit, SHAP feature attribution, LSTM comparison, and abstention confound testing—reveals that a three-feature HAR-style logistic regression matches or exceeds a 36-feature Random Forest at four of five prediction horizons under moving-block bootstrap testing. A coverage-matched persistence rule makes identical predictions to the Random Forest on every jointly covered day at horizons $N \geq 10$ (McNemar $b = c = 0$), confirming that any accuracy gap is attributable to coverage composition rather than multi-feature predictive skill. The confidence filter achieves 0% accuracy on calm-to-high volatility transition days at all horizons; explicit cost-weighting up to $20\times$ does not improve this, establishing an informational ceiling rather than an objective-function deficiency. These results replicate on S&P 500 trend regimes, establishing the persistence illusion as a structural property of selective prediction on serially correlated labels. The diagnostic protocol is modular and directly applicable to any selective prediction system on autocorrelated outcomes.

Keywords: VIX regime classification; selective prediction; persistence illusion; machine learning; Random Forest; volatility forecasting; autocorrelation artifact

JEL Classification: C53; C63; G10; G17; C12

1 Introduction

Volatility is one of the most important dimensions of financial market risk. The CBOE Volatility Index (VIX), often described as the market’s “fear gauge,” summarizes option-implied expectations of S&P 500 volatility over the next 30 days and is widely used by investors, risk managers, and policymakers to monitor uncertainty in equity markets [Whaley, 2009]. When VIX rises above 20—a level widely adopted as a practical divide between normal and elevated-fear environments [Whaley, 2009]—markets are generally considered to be in a high-volatility regime characterized by elevated fear, widening credit spreads, and increased probability of large equity drawdowns. Episodes such as the COVID-19 crash of March 2020 and the Japan carry-trade unwind of August 2024 illustrate the costs of failing to anticipate these regime transitions.

Despite its importance, VIX regime prediction is a difficult problem. Volatility is noisy, mean-reverting, and driven by a complex mixture of market sentiment, equity conditions, interest rates, and sudden exogenous shocks. Classical approaches such as GARCH-family models [Bollerslev, 1986] and Markov-switching frameworks [Hamilton, 1989] have provided important foundations, but often assume specific parametric structures that may not hold in practice. Machine learning methods offer flexibility in modeling nonlinear interactions across large feature sets, but introduce their own challenges: temporal leakage, calibration, and—critically for regime classification—the difficulty of distinguishing genuine predictive skill from the strong autocorrelation inherent in VIX.

The selective prediction literature—which studies classifiers that may abstain from uncertain predictions [Chow, 1970, Bartlett and Wegkamp, 2008, Geifman and El-Yaniv, 2017]—offers a principled solution: reserve predictions for high-confidence cases, trading coverage for reliability on predicted days. On autocorrelated label sequences such as VIX regimes, however, selective prediction can manufacture the appearance of skill by preferentially covering easy regime-stable days.

This paper develops a confidence-based selective prediction framework for VIX regime classification and uses it as a lens to examine where accuracy gains come from and what they are worth. Two ensemble classifiers—Random Forest (RF) and Histogram Gradient Boosting (HGB)—are trained on 36 features and abstain when calibrated probability is close to 0.5. The headline accuracy figures (90–93% for RF at $\tau = 0.25$) are striking, but a systematic

decomposition reveals that they constitute a diagnostic, not an achievement.

The main findings are as follows. First, a HAR-style logistic regression using only three features (current VIX level, 5-day VIX mean, 20-day VIX mean) matches or exceeds the full 36-feature RF in point estimate at four of five horizons at matched coverage; the HAR-minus-RF gap at $N = 10$ is +1.1 percentage points (pp; HAR leads), with bootstrap 95% CI $[-1.08, +4.18]$, so the advantage is a pattern, not a confirmed result. Second, a coverage-matched naive persistence rule closely tracks RF performance at all horizons; a McNemar test on jointly covered days finds $b = c = 0$ at $N \geq 10$: RF and persistence make identical predictions on every shared day. Third, the confidence filter covers 87% of calm-persisting days at $N = 5$ but yields 0% accuracy on every calm→high transition it covers; explicit cost-weighting up to $20\times$ does not overcome this. Fourth, at $N = 25$, balanced accuracy collapses to 50% for both RF and HGB. Fifth, persistence at matched coverage dominates RF selective prediction on cost-weighted utility whenever false negatives are more costly than false alarms ($C \geq 5$) at $N = 10$.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 describes the data. Section 4 presents the methodology. Section 5 reports empirical results. Section 6 replicates the diagnostic on S&P 500 trend regimes. Section 7 concludes.

2 Related Work

2.1 Volatility Forecasting and Regime Modeling

The modeling of financial volatility has a long history in econometrics. [Bollerslev \[1986\]](#) introduced the GARCH model, which captures the tendency of volatility to cluster over time by modeling conditional variance as a function of past squared residuals. GARCH-family models have since been extended in numerous directions and remain a standard benchmark for volatility forecasting. [Hamilton \[1989\]](#) introduced Markov-switching models, in which a time series is assumed to switch between discrete latent states according to a Markov chain. This framework provides a natural model for regime transitions and has been widely applied to financial and macroeconomic time series. [Guidolin and Timmermann \[2007\]](#) extended regime-switching ideas to multivariate portfolio allocation settings, demonstrating that regime-dependent models can outperform single-state specifications in out-of-sample asset allocation.

A complementary strand of work focuses on realized volatility constructed from high-frequency intraday returns. [Corsi \[2009\]](#) proposed the Heterogeneous Autoregressive (HAR) model, which captures the long-memory properties of realized volatility through a simple

additive structure of daily, weekly, and monthly volatility components. The HAR model has become a standard benchmark for realized volatility forecasting, and its strong performance motivates including a HAR-style logistic regression as a baseline in this paper.

In the VIX-specific literature, [Whaley \[2009\]](#) documented the VIX’s role as a fear indicator and its asymmetric behavior in market downturns. [Fernandes et al. \[2014\]](#) studied the predictability of VIX using a variety of econometric models and found that simple autoregressive specifications perform competitively against more complex alternatives. This paper replicates and extends that finding: simple HAR-style models are competitive with 36-feature ensemble classifiers on the binary regime classification task.

2.2 Machine Learning for Financial Prediction

Machine learning methods have increasingly been applied to financial forecasting. Ensemble methods such as Random Forests [[Breiman, 2001](#)] and Gradient Boosting are particularly well-suited to financial prediction because they can model nonlinear interactions among large feature sets without requiring distributional assumptions.

A key methodological challenge in applying machine learning to financial data is avoiding data leakage and respecting the temporal ordering of observations. This paper addresses this by using chronological train/test splitting, `TimeSeriesSplit` cross-validation during calibration, and walk-forward validation as a robustness check—consistent with best practices in applied financial machine learning.

2.3 Selective Prediction and the Reject Option

The reject option in classification allows a model to abstain from making a prediction when its confidence is insufficient. [Chow \[1970\]](#) first formalized the optimal reject option. [Bartlett and Wegkamp \[2008\]](#) studied classification with a reject option using hinge loss and established theoretical generalization bounds. [Geifman and El-Yaniv \[2017\]](#) demonstrated that selective prediction can substantially improve reliability in modern deep learning settings.

In financial applications, the reject option is particularly natural. A risk model that provides no signal on ambiguous days is often more useful than one that forces a low-confidence prediction. However, on autocorrelated series such as VIX regimes, abstention can manufacture accuracy without genuine predictive skill—the central diagnostic this paper examines.

3 Data

3.1 Market Data

Daily price data are obtained from Yahoo Finance for eight market series: the CBOE Volatility Index (VIX), the 3-month VIX term structure index (VIX3M), the CBOE 9-Day Volatility Index (VIX9D), the S&P 500 Index, gold futures (GC=F), the 10-Year Treasury Yield (TNX), the U.S. Dollar Index (DXY), and the CBOE SKEW Index. VIX9D is downloaded to keep the raw data complete but is not used as a model feature. Although raw VIX data is available from January 1990, VIX3M data begins in July 2006; because VIX3M anchors the term spread feature, the effective sample starts on July 17, 2006.

3.2 Macroeconomic Data

Two macroeconomic variables are obtained from the Federal Reserve Economic Data (FRED) API: the Federal Funds Rate and the 10-Year minus 2-Year Treasury yield curve spread (T10Y2Y). The Federal Funds Rate target is forward-filled to daily frequency by holding the prevailing rate constant between FOMC meeting dates.

3.3 Label Construction

The prediction task is binary regime classification. For a prediction horizon N , the label is:

$$\text{LABEL}_{N,t} = \begin{cases} 1 & \text{if } \text{VIX}_{t+N} \geq 20 \\ 0 & \text{if } \text{VIX}_{t+N} < 20 \end{cases} \quad (1)$$

The label is forward-shifted N days to ensure that all features are observed at time t and the outcome is measured at time $t+N$, eliminating lookahead bias. Horizons $N \in \{5, 10, 15, 20, 25\}$ trading days are evaluated.

A *sustained-high* label is also constructed:

$$\text{SUSTAINED_LABEL}_{N,t} = \begin{cases} 1 & \text{if } \text{VIX}_{t+k} \geq 20 \text{ for all } k \in \{1, \dots, N\} \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

3.4 Class Distribution and Train/Test Split

The full feature dataset covers 5,000 trading days from July 17, 2006 to May 29, 2026. Data are split chronologically, with the first 80% used for training (2006–2022) and the remaining

20% for testing (2022–2026), preserving strict temporal ordering.

Table 1: Class distribution (fraction of days with $VIX \geq 20$) by dataset split.

Split	$VIX \geq 20$	$VIX < 20$	Majority-class baseline
Training (2006–2022)	36.7%	63.3%	63.3%
Test (2022–2026)	29.8%	70.2%	70.2%

High-volatility days are substantially less common in the test period ($\approx 30\%$) than in training ($\approx 37\%$), reflecting that the training window encompasses several high-volatility crises—the 2008–09 financial collapse, the 2011 eurozone episode, and the COVID-19 crash—whereas the test window is dominated by the post-bear-market recovery despite the elevated volatility of the first half of 2022.

4 Methodology

4.1 Feature Engineering

The model uses 36 engineered features grouped into five categories. All features are computed using only information available at or before the prediction date t .

VIX technical indicators (13 features): 10-day and 20-day simple moving averages (SMA10, SMA20); 10-day and 20-day momentum; 10-day and 20-day rate of change; Bollinger Band standard deviation, upper and lower bands, and position within the band; 14-day RSI; moving-average cross (SMA10 – SMA20); and VIX term spread ($VIX3M - VIX$).

SKEW features (3 features): SKEW level, 5-day change, and 20-day change.

S&P 500 features (7 features): 1-day, 5-day, and 20-day returns; 10-day and 20-day price momentum; 20-day realized volatility; and 252-day maximum drawdown.

Cross-asset features (9 features): Gold 1-day, 5-day, and 20-day returns; 10-Year Treasury Yield 1-day, 5-day, and 20-day changes; DXY 1-day, 5-day, and 20-day returns.

Macroeconomic features (4 features): Federal Funds Rate 1-month and 3-month changes; yield curve spread 1-month and 3-month changes.

The complete feature list appears in Appendix A.

4.2 Models and Training Pipeline

Two ensemble classifiers are evaluated: Random Forest (RF) and Histogram Gradient Boosting (HGB), both implemented via scikit-learn. Both use a single fixed specification: RF

with `n_estimators=200`, `max_depth=10`, `min_samples_leaf=5`; HGB with `max_iter=200`, `max_depth=5`, `learning_rate=0.1`. All seeds set to `random.state=42`. Using one specification eliminates pipeline inconsistencies and provides a fair comparison across tables. All features are standardized using `StandardScaler` fit on the training set and applied to the test set.

4.3 Probability Calibration

The selective prediction rule requires well-calibrated probability estimates. We apply isotonic regression calibration using `CalibratedClassifierCV` with 5-fold `TimeSeriesSplit` cross-validation on the full training set. The calibrated probability at time t is denoted $\hat{p}_t$, representing the estimated probability that VIX will be in a high-volatility regime at $t + N$.

4.4 Confidence-Based Selective Prediction

The selective classifier makes a prediction only when $\hat{p}_t$ is sufficiently far from 0.5. For confidence threshold $\tau \in (0, 0.5)$, the prediction rule is:

$$\hat{y}_t = \begin{cases} 1 & \text{if } \hat{p}_t > 0.5 + \tau \\ 0 & \text{if } \hat{p}_t < 0.5 - \tau \\ \text{abstain} & \text{if } 0.5 - \tau \leq \hat{p}_t \leq 0.5 + \tau \end{cases} \quad (3)$$

Accuracy is evaluated only on days where the model makes a prediction. Coverage is defined as the fraction of test days on which a prediction is made. The primary threshold is $\tau = 0.25$.

5 Results

5.1 Main Accuracy Results

Table 2 reports baseline and selective accuracy, coverage, and balanced accuracy on covered days at $\tau = 0.25$.

Table 2: Baseline and selective ($\tau = 0.25$) accuracy, coverage, and balanced accuracy on covered days. Gain = selective minus baseline in percentage points (pp).

N	RF Base	RF Sel	RF Gain	RF Cov	RF BalAcc	HGB Sel	HGB Gain	HGB Cov	HGB BalAcc
5	83.4%	91.9%	+8.5 pp	76.3%	88.4%	92.1%	+8.9 pp	71.1%	88.2%
10	79.5%	90.0%	+10.5 pp	60.1%	84.8%	92.2%	+13.6 pp	63.1%	86.1%
15	76.2%	90.0%	+13.8 pp	51.3%	83.9%	92.2%	+21.3 pp	36.0%	89.5%
20	76.8%	92.7%	+15.9 pp	31.7%	79.1%	94.8%	+22.2 pp	26.8%	84.1%
25	73.4%	90.3%	+16.9 pp	26.9%	50.0%	92.7%	+26.5 pp	19.2%	50.0%

Table 3: Extended metrics on covered days for Random Forest ($\tau = 0.25$). At $N = 25$, precision and recall are both 0% and balanced accuracy equals 50%, confirming the model exclusively predicts the calm class at this horizon.

N	Selective Acc	Precision	Recall	F1	Balanced Acc
5	91.9%	86.5%	81.2%	83.8%	88.4%
10	90.0%	86.8%	73.8%	79.7%	84.8%
15	90.0%	89.8%	70.8%	79.2%	83.9%
20	92.7%	100.0%	58.2%	73.6%	79.1%
25	90.3%	0.0%	0.0%	0.0%	50.0%

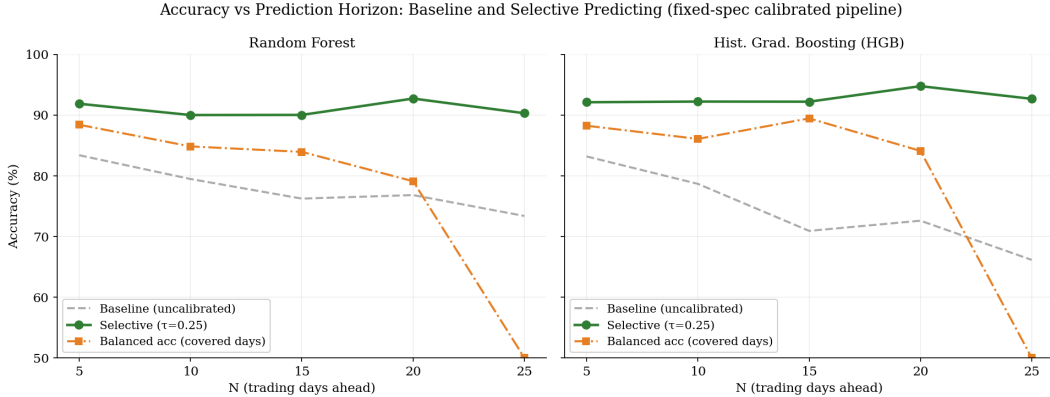


Figure 1: Baseline, selective accuracy, and balanced accuracy across prediction horizons for Random Forest (left) and HGB (right). At $N = 25$, balanced accuracy equals 50%, confirming a coin-flip result despite 90% selective accuracy.

Selective prediction improves accuracy for both models at every horizon. However, the balanced accuracy columns reveal the underlying artifact: RF balanced accuracy declines monotonically from 88.4% at $N = 5$ to 50.0% at $N = 25$. This collapse confirms that

the model exclusively predicts calm outcomes at the longest horizon, and the nominal 90% selective accuracy reflects the 70% base rate of calm days, not predictive skill.

5.2 Threshold Sensitivity

Table 4: Random Forest selective accuracy and coverage as a function of confidence threshold τ .

τ	RF $N=5$ Acc	$N=5$ Cov	RF $N=10$ Acc	$N=10$ Cov
0.10	86.5%	91.0%	83.3%	90.5%
0.15	88.3%	85.1%	84.7%	85.9%
0.20	89.0%	82.9%	86.6%	78.5%
0.25	91.9%	76.3%	90.0%	60.1%
0.30	92.3%	73.6%	91.8%	53.6%
0.35	95.0%	67.1%	95.1%	36.5%
0.40	97.4%	49.0%	97.4%	15.3%

Accuracy increases monotonically with τ across all horizons, confirming that the calibrated probability is a reliable ranking signal. However, as Section 5.7 shows, increasing τ does not improve coverage of transition days—it makes confident wrong predictions on calm→high days even more selective.

5.3 Robustness Across VIX Thresholds

Table 5: Random Forest accuracy gain (selective vs. baseline) by VIX regime threshold at $\tau = 0.25$.

N	VIX ≥ 18	VIX ≥ 20	VIX ≥ 22
5	+5.4 pp	+8.5 pp	+7.5 pp
10	+11.3 pp	+10.5 pp	+10.3 pp
15	+14.8 pp	+13.8 pp	+7.9 pp
20	+22.3 pp	+15.9 pp	+9.1 pp
25	−14.8 pp	+16.9 pp	+8.3 pp

The advantage of selective prediction holds across VIX thresholds of 18, 20, and 22 at $N \leq 20$. The exception is RF at $N = 25$, VIX ≥ 18 , which shows a gain of −14.8 pp, consistent with the degenerate behavior documented in Table 3.

5.4 Sustained Regime Labels

Table 6: Random Forest selective accuracy for instantaneous and sustained regime labels at $\tau = 0.25$.

N	Instantaneous	Sustained
5	91.9%	94.0%
10	90.0%	93.8%
15	90.0%	93.8%
20	92.7%	94.3%
25	90.3%	89.9%

Sustained regime labels are consistently easier to predict than instantaneous labels at $N \leq 20$, reflecting that persistent high-volatility episodes are preceded by stronger signals across multiple features.

5.5 Comparison to Naive Baselines

A critical question is whether the RF selective accuracy reflects genuine multi-feature ML skill or simply VIX autocorrelation. To answer this, we construct four coverage-matched baselines evaluated on the same test set.

The **persistence baseline** predicts $VIX_{t+N} \geq 20$ if and only if $VIX_t \geq 20$, and abstains when $|VIX_t - 20| < c$ for a constant c chosen to match the RF’s coverage. The **LR (VIX only) baseline** applies the full calibration pipeline to a single feature (current VIX level). The **HAR-style logistic regression** uses three features motivated by the HAR model [Corsi, 2009]: current VIX (VIX_D), 5-day VIX mean (VIX_W), and 20-day VIX mean (VIX_M). The **Markov-chain baseline** estimates the 1-step transition probability matrix from training data and raises it to the N th power.

Table 7: RF selective accuracy vs. naive baselines at matched coverage ($\tau = 0.25$). Bolded values indicate where a baseline exceeds the RF selective accuracy.

N	Coverage	Persistence	LR (VIX)	HAR-LR	Markov	RF Sel	RF–Persist
5	76.3%	91.6%	91.8%	91.6%	83.7%	91.9%	+0.3 pp
10	60.1%	88.1%	87.6%	91.1%	85.3%	90.0%	+1.9 pp
15	51.3%	87.9%	88.4%	90.2%	84.8%	90.0%	+2.1 pp
20	31.7%	90.5%	87.9%	94.2%	84.5%	92.7%	+2.3 pp
25	26.9%	83.8%	86.8%	92.0%	82.9%	90.3%	+6.5 pp

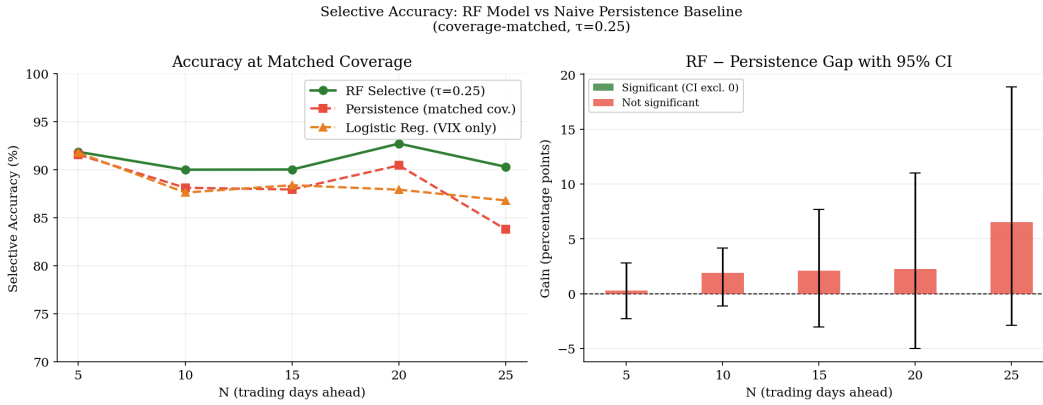


Figure 2: Left: RF selective vs. naive baseline accuracy at matched coverage across horizons. Right: RF vs. persistence gap with 95% moving-block bootstrap CI. No bar is statistically significant at 5%.

Table 7 delivers the paper’s sharpest negative result. The HAR-style logistic regression—using only three VIX-related features—matches or exceeds the full 36-feature RF in point estimate at four of five horizons ($N = 10, 15, 20$, and 25). A moving-block bootstrap on the HAR-minus-RF gap at $N = 10$ (HAR leads by +1.1 pp) yields 95% CI $[-1.08, +4.18]$; the corresponding RF-minus-persistence bootstrap gives CI $[-1.11, +4.17]$. Neither gap is statistically confirmed. The 33 additional features contribute negligibly beyond the three-variable VIX autoregressive structure.

5.6 Persistence Quantification

For each horizon N , we compute the empirical transition probabilities $\Pr(\text{VIX}_{t+N} \geq 20 \mid \text{VIX}_t \geq 20)$ and $\Pr(\text{VIX}_{t+N} \geq 20 \mid \text{VIX}_t < 20)$, and classify each correct RF covered prediction as either a same-regime or regime-change call.

Table 8: Empirical VIX transition probabilities and decomposition of RF correct covered predictions. $P(H|H) = P(\text{VIX}_{t+N} \geq 20 \mid \text{VIX}_t \geq 20)$; $P(H|C) = P(\text{VIX}_{t+N} \geq 20 \mid \text{VIX}_t < 20)$. Same-regime % = fraction of correct RF covered predictions that match the regime at t .

N	$P(H H)$	$P(H C)$	Persist. ratio	RF cov. acc	Same-regime %
5	84.2%	8.6%	9.8×	91.9%	99.0%
10	79.5%	11.2%	7.1×	90.0%	99.6%
15	75.9%	13.2%	5.8×	90.0%	99.8%
20	73.1%	14.8%	4.9×	92.7%	100.0%
25	70.0%	16.5%	4.2×	90.3%	100.0%

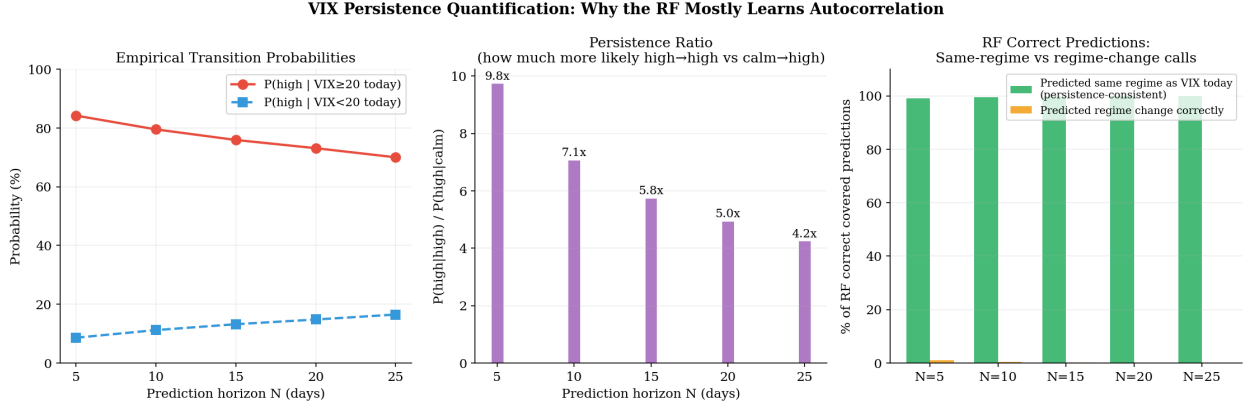


Figure 3: Left: empirical transition probabilities conditional on today’s regime. Center: persistence ratio $P(H|H)/P(H|C)$. Right: decomposition of RF correct covered predictions into same-regime and regime-change calls. At $N \geq 20$, 100% of correct predictions are same-regime.

At $N = 5$, the probability of VIX being high in 5 days is 84.2% given it is high today, compared to only 8.6% given it is calm today—a 9.8× ratio. On covered days, 99.0% of the RF’s correct predictions at $N = 5$ are same-regime calls; at $N \geq 20$, every correct covered prediction is a persistence call. The RF adds no directional forecasting beyond VIX autocorrelation on the days it chooses to cover.

5.7 Regime Transition Analysis

We classify each test day by the regime at time t and the realized regime at $t + N$, creating four transition categories: calm→calm, calm→high, high→calm, and high→high.

Table 9: RF selective coverage and accuracy by regime transition type ($\tau = 0.25$, test set). Calm→high accuracy is **0%** at all horizons: every covered prediction on a day about to spike is a confident “stay calm” signal.

Transition	$N = 5$			$N = 10$		
	Days	Coverage	Acc (covered)	Days	Coverage	Acc (covered)
calm → calm	614	87.1%	99.6%	587	71.6%	100.0%
calm → high	79	45.6%	0.0%	101	41.6%	0.0%
high → calm	84	35.7%	23.3%	111	18.0%	10.0%
high → high	222	72.5%	99.4%	199	59.3%	100.0%

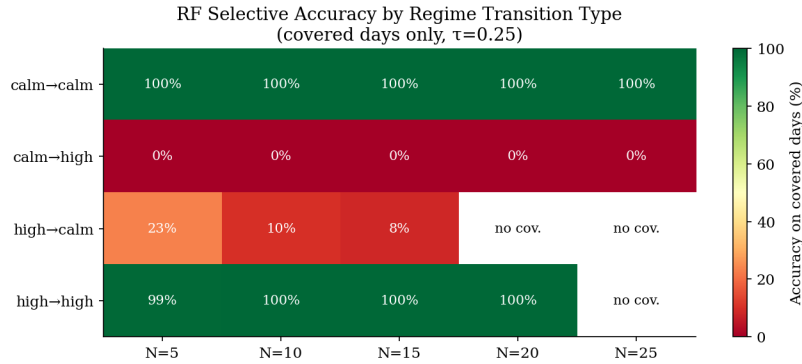


Figure 4: RF selective accuracy by regime transition type and horizon (covered days only, $\tau = 0.25$). Calm→high rows are uniformly 0% at all horizons.

At $N = 5$, the model covers 36 of 79 calm→high days (45.6%). On every single one of those 36 days, it issues a confident “stay calm” prediction and is wrong. The mechanism is clear: VIX comfortably below 20 generates calibrated probabilities below 0.25, triggering a high-confidence calm prediction.

Structural caveat. This failure is not a model-specific defect: it is an expected structural property of the prediction problem. Calm→high transitions are disproportionately triggered by exogenous events—geopolitical shocks, central bank surprises, sudden liquidity dislocations—whose onset leaves no advance signature in publicly available daily price and macro features. The diagnostic value of 0% covered accuracy is that the confidence filter fails to surface its own ignorance, issuing confident wrong predictions on covered transition days rather than abstaining.

Event case studies. Figure 5 documents the model’s behavior during four major volatility events in the test set: the 2022 bear-market VIX peak (June 2022), the UK gilt crisis (October 2022), the SVB banking collapse (March 2023), and the Japan carry-trade unwind (August 2024, when VIX reached 65).

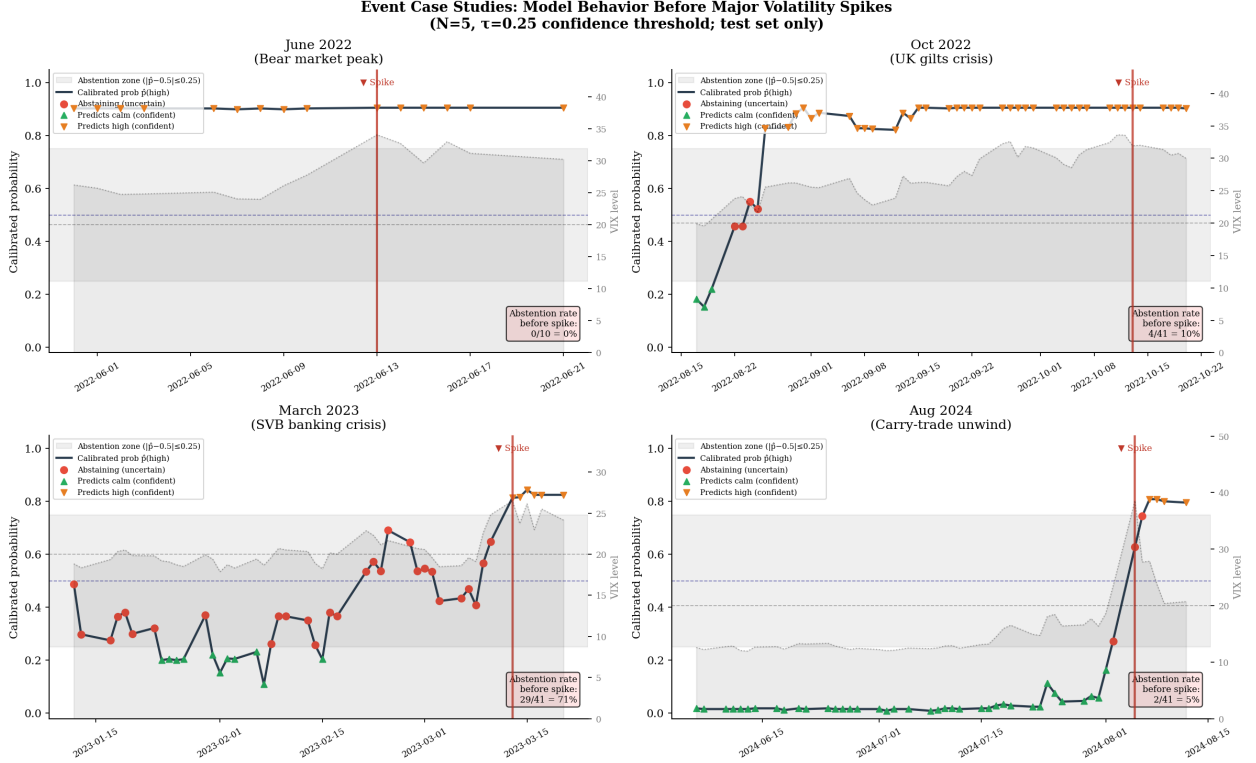


Figure 5: Model behavior ($\hat{p}_t$, $N = 5$, $\tau = 0.25$) for 40 trading days before four major volatility events. In all four cases the model issues confident calm predictions leading up to the event.

Across all four events the pattern is consistent: the model issues confident “stay calm” predictions on the days leading up to each spike. The August 2024 carry-trade unwind shows essentially no anticipatory signal; $\hat{p}_t$ remains below $0.5 - \tau$ until the day the spike materializes.

Cost-weighting does not rescue calm→high recall. To confirm the failure is informational rather than objective-function-driven, we refit the RF with explicit class weights $\{0:1, 1:w\}$ for $w \in \{1, 5, 10, 20\}$.

Table 10: Effect of class-weighting on transition coverage and selective accuracy. Calm→high accuracy is **0.0%** at every weight for both horizons.

N	w	C→H Cov	C→H Acc	C→C Cov	C→C Acc	Sel Acc
5	1	45.6%	0.0%	87.1%	99.6%	91.9%
5	5	46.8%	0.0%	87.1%	100.0%	92.2%
5	10	49.4%	0.0%	87.3%	100.0%	92.3%
5	20	48.1%	0.0%	84.4%	100.0%	91.8%
10	1	41.6%	0.0%	71.6%	100.0%	90.0%
10	5	49.5%	0.0%	77.7%	100.0%	89.5%
10	10	58.4%	0.0%	80.4%	100.0%	87.4%
10	20	57.4%	0.0%	82.8%	100.0%	87.4%

The failure to predict calm→high transitions is *informational*, not objective-function-driven.

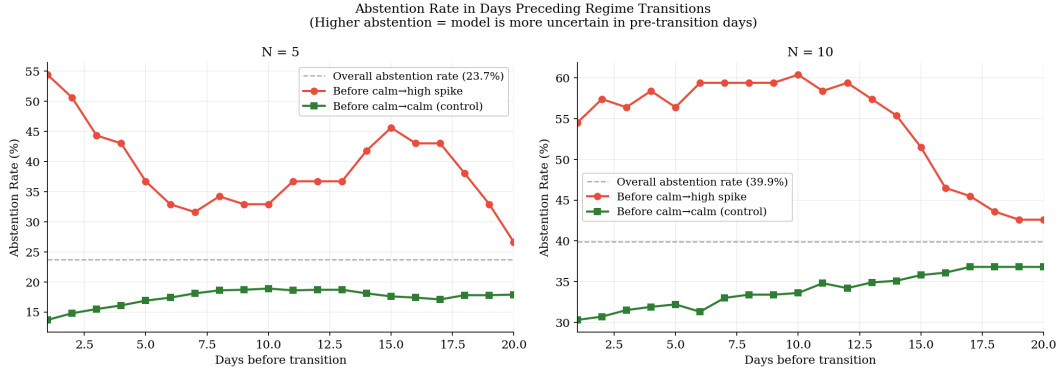


Figure 6: Abstention rate before calm→high transitions (red) vs. calm→calm days (green). At lag = 1 day, the abstention rate before spikes is more than double the overall rate.

Abstention as a leading indicator. At $N = 5$, the abstention rate immediately before a calm→high transition is 54.4%, compared to 13.7% before calm→calm days and 23.7% overall—a $2.3\times$ elevation. However, a confound test shows that VIX proximity to the threshold ($\text{AUC} = 0.805$) explains most of this signal; the abstention indicator alone achieves $\text{AUC} = 0.708$. At $N = 5$, abstention carries a small residual signal beyond proximity (likelihood-ratio $p = 0.001$ on the evaluation sample, treated as descriptive rather than confirmatory). This observation should be treated as hypothesis-generating and requires dedicated out-of-sample confirmation.

5.8 Walk-Forward Validation

Table 11: 5-fold expanding walk-forward validation. Shaded rows indicate that at $N = 20$ the RF/HGB standard deviations exceed 25 pp; persistence remains stable.

Model	N	Mean Baseline	Std	Mean Selective	Std
Random Forest	5	82.8%	9.7%	92.2%	1.7%
Persistence	5	88.0%	3.2%	93.9%	3.2%
HGB	5	84.0%	8.0%	84.6%	9.7%
Random Forest	10	72.3%	22.7%	85.9%	7.7%
Persistence	10	84.8%	4.5%	91.3%	4.1%
HGB	10	70.9%	22.9%	77.3%	15.2%
Random Forest	20	67.5%	22.8%	67.1%	28.3%
Persistence	20	81.5%	3.9%	86.2%	7.7%
HGB	20	63.6%	25.2%	61.4%	29.9%

Coverage-matched persistence at $N = 5$ achieves $93.9\% \pm 3.2\%$ —higher mean than RF ($92.2\% \pm 1.7\%$) with roughly double the fold-to-fold variance. At $N = 10$, persistence ($91.3\% \pm 4.1\%$) exceeds RF selective ($85.9\% \pm 7.7\%$). At $N = 20$, both RF and HGB show standard deviations exceeding 25 pp, while persistence remains stable. The practical reliable prediction window is approximately $N \leq 10$.

5.9 LSTM Comparison

A two-layer LSTM network is trained on sliding windows of the past 20 trading days of all 36 features, using the same 80/20 chronological split. The LSTM uses hidden size 64, dropout 0.3, and is trained for 40 epochs with the Adam optimizer.

Table 12: LSTM vs. RF at $\tau = 0.25$. BalAcc = balanced accuracy on covered days.

N	LSTM Sel Acc	LSTM Cov	LSTM BalAcc	RF Sel Acc	RF Cov
5	76.5%	87.0%	76.0%	91.7%	77.0%
10	71.8%	90.7%	64.7%	89.3%	64.3%

The LSTM achieves non-zero calm→high covered accuracy (34.9% at $N = 5$; 19.1% at $N = 10$)—unlike the RF’s 0%—but both figures are substantially below the 50% random

baseline, meaning the LSTM is wrong more often than right on transition days. Architectural richness (20-day temporal memory) does not resolve the information constraint: the daily feature set does not carry the event-anticipation content needed to forecast volatility spikes.

5.10 SHAP Feature Attribution

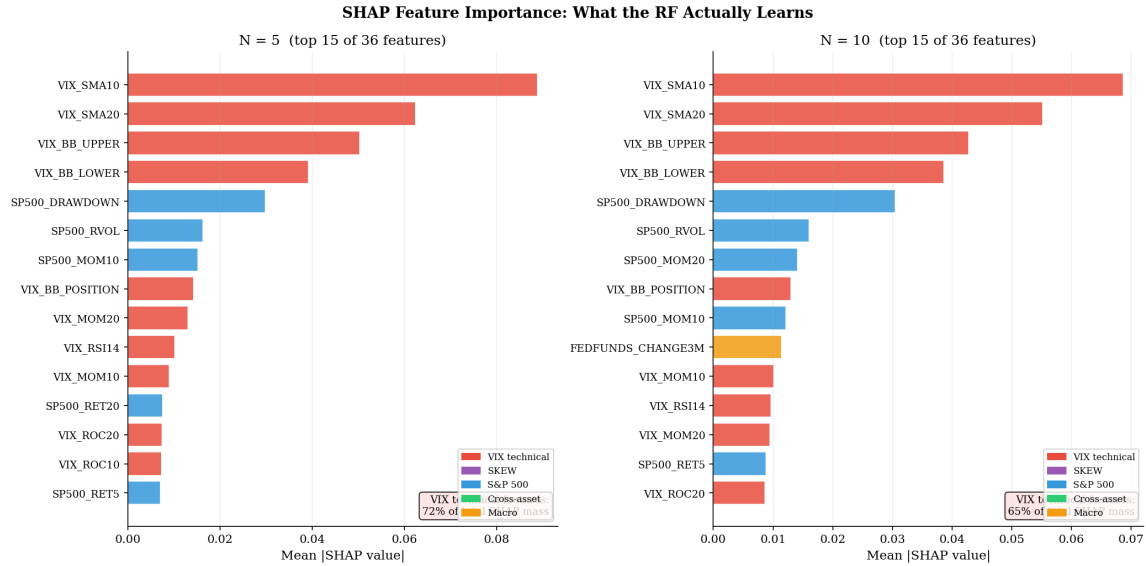


Figure 7: Mean absolute SHAP values (top 15 of 36 features) at $N = 5$ (left) and $N = 10$ (right). VIX technical features account for 71.9% and 64.7% of total SHAP mass, respectively.

VIX technical features account for **71.9%** of total SHAP mass at $N = 5$ and **64.7%** at $N = 10$. The dominant features are all smoothed or envelope transforms of the current VIX level. SHAP confirms that the model’s predictions are driven overwhelmingly by VIX autocorrelation, with negligible contribution from the broader cross-asset and macro feature set.

5.11 Statistical Significance of the Persistence Gap

We apply a moving-block bootstrap to the paired accuracy difference (RF selective minus persistence, at matched coverage). Block length is set to $\max(\sqrt{N_{\text{test}}}, 20) \approx 32$ days [Kunsch, 1989]; 2,000 bootstrap replicates are drawn.

Table 13: Moving-block bootstrap 95% confidence intervals for the RF vs. persistence accuracy gap. All CIs include 0; no gap is statistically significant at the 5% level.

N	Gap (pp)	95% CI lower	95% CI upper
5	+0.30	−2.26	+2.80
10	+1.88	−1.11	+4.17
15	+2.07	−3.03	+7.70
20	+2.28	−5.01	+11.03
25	+6.52	−2.86	+18.87

No gap is statistically significant at the 5% level. The McNemar test sharpens this: at $N \geq 10$, RF and persistence make *identical* predictions on every jointly covered day ($b = c = 0$, $\chi^2 = 0$, $p = 1.000$). The Table 13 accuracy gaps are therefore attributable entirely to differences in which days each rule covers, not to differing predictions on shared days. Excluding the 2022 bear-market period (149–165 test days) leaves all qualitative conclusions unchanged: calm→high covered accuracy remains 0% at all horizons.

5.12 Cost-Weighted Utility Analysis

$$U = -\frac{\text{FP} + C \cdot \text{FN}}{n_{\text{predictions}}} \quad (4)$$

Table 14: Cost-weighted utility per prediction under different FN/FP cost ratios C . Higher utility (closer to 0) is better. Bolded entries show where persistence dominates RF selective.

N	C	RF Selective	Persistence	RF Baseline	Always Calm
5	1	−0.081	−0.087	−0.166	−0.301
5	5	−0.276	−0.286	−0.515	−1.507
5	10	−0.518	−0.535	−0.950	−3.013
5	20	−1.004	−1.033	−1.821	−6.026
10	1	−0.100	−0.120	−0.205	−0.301
10	5	−0.380	− 0.326	−0.626	−1.503
10	10	−0.730	− 0.584	−1.152	−3.006
10	20	−1.430	− 1.100	−2.204	−6.012

At $N = 10$, persistence at matched coverage dominates RF selective prediction under any moderate false-negative cost asymmetry ($C \geq 5$). The RF selective framework is

utility-beneficial only when false positives are as costly as false negatives ($C = 1$).

6 Replication: S&P 500 Trend-Regime Classification

To test whether the persistence illusion is specific to volatility forecasting, we replicate the full diagnostic protocol on S&P 500 trend-regime classification using the same feature set and model specification. The label is:

$$\text{BEAR}_{N,t} = \begin{cases} 1 & \text{if } \text{SP500}_{t+N} < \overline{\text{SP500}}_{200,t+N} \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

Table 15: S&P 500 trend-regime replication. RF–Persist is *negative* at every horizon: persistence strictly dominates the RF selective classifier.

N	Base rate	Coverage	Persistence	HAR-analog	RF Sel	RF–Persist
5	22.0%	74.4%	100.0%	86.9%	96.0%	−4.0 pp
10	21.9%	68.3%	99.6%	86.4%	94.9%	−4.7 pp
15	21.9%	69.7%	97.3%	86.7%	90.9%	−6.3 pp
20	21.8%	70.3%	94.9%	84.9%	87.7%	−7.1 pp
25	21.7%	59.5%	94.8%	84.8%	88.3%	−6.4 pp

Table 16: S&P 500 transition accuracy. Bull→bear covered accuracy is **0%**—exact mirror of VIX calm→high.

N	Transition	Days	Coverage	Acc (covered)	Same-regime %
5	bull→bull	749	88.8%	100.0%	99.2%
5	bull→bear	25	68.0%	0.0%	
10	bull→bull	730	86.2%	100.0%	98.6%
10	bull→bear	39	71.8%	0.0%	

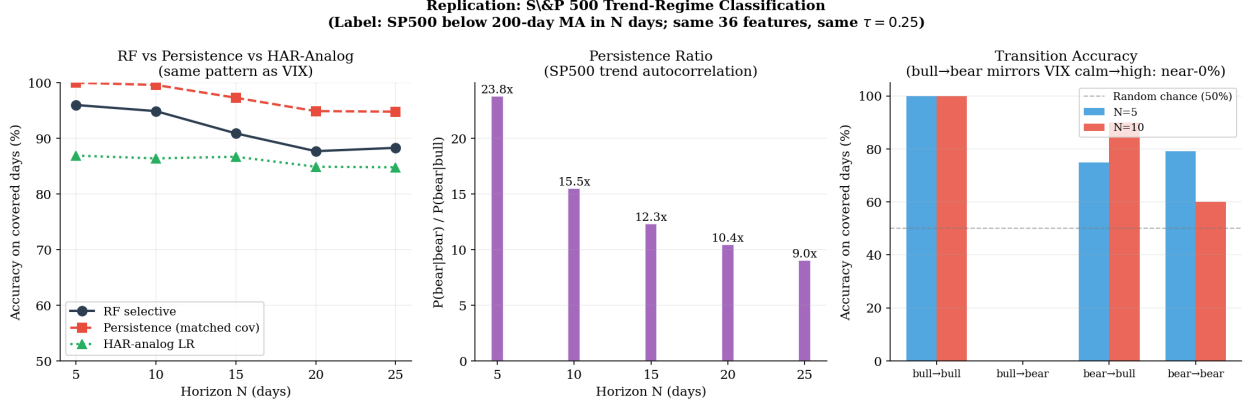


Figure 8: S&P 500 trend-regime replication. Left: RF vs. persistence—persistence dominates at all horizons. Center: persistence ratio of $23.8\times$ at $N = 5$. Right: bull→bear covered accuracy is 0%, replicating the VIX calm→high finding.

The persistence ratio is **$23.8\times$** at $N = 5$ —more than twice the VIX ratio of $9.7\times$ —and persistence beats the RF at *every horizon* (RF–Persist: -4.0 to -7.1 pp). Bull→bear covered accuracy is **0.0%** at both $N = 5$ and $N = 10$, replicating the VIX calm→high finding exactly. The replication establishes that the persistence illusion is a structural consequence of applying selective prediction to any autocorrelated binary time series, not a VIX-specific artifact.

7 Conclusions

This paper applies a six-component diagnostic protocol to VIX regime classification and shows that headline selective prediction accuracy reflects autocorrelation, not machine learning skill. The confidence filter achieves 0% accuracy on every calm→high transition it covers; explicit cost-weighting up to $20\times$ does not rescue this. A McNemar test on jointly covered days finds $b = c = 0$ at $N \geq 10$: RF and persistence make identical predictions on every shared day, so the accuracy gap is entirely composition-driven. Three-feature HAR logistic regression matches or exceeds 36-feature RF in point estimate at four of five horizons; cost-weighted utility at $N = 10$ favors persistence whenever false negatives are more costly than false alarms.

SHAP attribution shows VIX technical features account for 71.9% of model prediction mass; a two-layer LSTM also fails on calm→high transitions ($19\text{--}35\%$ accuracy, below random chance); and the full diagnostic replicates on S&P 500 trend regimes with bull→bear covered accuracy of 0% .

Limitations

The feature set consists of publicly available daily market data. Calm→high transitions are disproportionately triggered by exogenous shocks—geopolitical events, policy surprises, sudden liquidity crises—that leave no advance signature in daily price features. The four-year test period ($\approx 1,000$ trading days) is underpowered for distinguishing accuracy differences of 1–3 pp under serial dependence.

Future Work

The primary open direction is feature enrichment: incorporating event-anticipation content such as options skew dynamics, credit spreads, or order-flow imbalance. A two-stage system using the abstention rate as a leading indicator followed by a conditional spike-prediction model is a second natural direction. Third, the diagnostic protocol should be applied to additional autocorrelated regime classification tasks—credit spread regimes, recession prediction, currency crisis indicators—to characterize how the persistence ratio relates to the severity of the selective-prediction bias.

Author Contributions

Conceptualization, A.G.; methodology, A.G.; software, A.G.; formal analysis, A.G.; data curation, A.G.; investigation, A.G.; writing—original draft, A.G.; writing—review and editing, J.L.; visualization, A.G.; supervision, J.L. All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement

The data used in this study are publicly available. VIX, VIX3M, S&P 500, gold, Treasury yield, DXY, and SKEW data are available from Yahoo Finance (<https://finance.yahoo.com>). Macroeconomic data (Federal Funds Rate and yield curve spread) are available from the Federal Reserve Economic Data (FRED) API (<https://fred.stlouisfed.org>).

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration of Generative AI Use

During the preparation of this work, the authors used Claude (Anthropic) to critique earlier manuscript drafts and improve writing clarity. The authors reviewed and edited the output as needed and take full responsibility for the content of the published article.

A Complete Feature List

Table 17: All 36 engineered features, grouped by category. All features are computed from information available at or before prediction date t .

#	Feature name	Definition	Category
1	VIX_SMA10	10-day simple moving average of VIX	VIX trend
2	VIX_SMA20	20-day simple moving average of VIX	VIX trend
3	VIX_MOM10	$VIX_t - VIX_{t-10}$	VIX momentum
4	VIX_MOM20	$VIX_t - VIX_{t-20}$	VIX momentum
5	VIX_ROC10	$(VIX_t/VIX_{t-10}) - 1$	VIX momentum
6	VIX_ROC20	$(VIX_t/VIX_{t-20}) - 1$	VIX momentum
7	VIX_BB.STD	20-day rolling std. of VIX	VIX Bollinger
8	VIX_BB.UPPER	$VIX_SMA20 + 2 \times VIX_BB_STD$	VIX Bollinger
9	VIX_BB.LOWER	$VIX_SMA20 - 2 \times VIX_BB_STD$	VIX Bollinger
10	VIX_BB.POSITION	$(VIX_t - VIX_BB_LOWER) / (4 \times VIX_BB_STD)$	VIX Bollinger
11	VIX_RSI14	14-day Relative Strength Index of VIX	VIX oscillator
12	VIX_MACROSS	$VIX_SMA10 - VIX_SMA20$	VIX trend
13	VIX_TERM.SPREAD	$VIX3M_t - VIX_t$	VIX structure
14	SKEW_LEVEL	Raw CBOE SKEW index value	Tail risk
15	SKEW_CHG5	$SKEW_t - SKEW_{t-5}$	Tail risk
16	SKEW_CHG20	$SKEW_t - SKEW_{t-20}$	Tail risk
17	SP500_RET1	Log return of S&P 500 over 1 day	Equity
18	SP500_RET5	Log return of S&P 500 over 5 days	Equity
19	SP500_RET20	Log return of S&P 500 over 20 days	Equity
20	SP500_MOM10	S&P 500 price at t - price at $t - 10$	Equity
21	SP500_MOM20	S&P 500 price at t - price at $t - 20$	Equity
22	SP500_RVOL	20-day realized volatility of S&P 500 daily returns	Equity
23	SP500_DRAWDOWN	252-day maximum drawdown of S&P 500	Equity
24	GOLD_RET1	Log return of gold futures over 1 day	Cross-asset
25	GOLD_RET5	Log return of gold futures over 5 days	Cross-asset
26	GOLD_RET20	Log return of gold futures over 20 days	Cross-asset
27	TNY_CHG1	10Y Treasury yield change over 1 day	Cross-asset
28	TNY_CHG5	10Y Treasury yield change over 5 days	Cross-asset
29	TNY_CHG20	10Y Treasury yield change over 20 days	Cross-asset
30	DXY_RET1	Log return of DXY over 1 day	Cross-asset
31	DXY_RET5	Log return of DXY over 5 days	Cross-asset
32	DXY_RET20	Log return of DXY over 20 days	Cross-asset
33	FEDFUNDS_CHG1M	Federal Funds Rate change over ≈ 21 trading days	Macro
34	FEDFUNDS_CHG3M	Federal Funds Rate change over ≈ 63 trading days	Macro
35	YIELD_CURVE_CHG1M	T10Y2Y spread change over ≈ 21 trading days	Macro
36	YIELD_CURVE_CHG3M	T10Y2Y spread change over ≈ 63 trading days	Macro

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